Teng Zhang

CURRICULUM VITAE

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EDUCATION

Sep. 2021–Present **Xi'an Jiaotong University, Xi'an, China**
Ph.D. student in Mathematics, School of Mathematics and Statistics.
Advisor: Zongben Xu. Previously studied under the guidance of Minghua Lin.

Dec. 2025–Dec. 2026 **University of Szeged, Szeged, Hungary**
Joint Ph.D. student in Mathematics, Department of Analysis, Bolyai Institute.
Advisor: Lajos Molnár.

Sep. 2017–Jul. 2021 **Taiyuan University of Technology, Taiyuan, China**
B.S. in Mathematics, School of Mathematics.

RESEARCH INTERESTS

Complex analysis; operator algebra; matrix analysis.

RESEARCH FUNDING

Jan. 2025–Dec. 2026 **CAST Young Talent Support Program (Doctoral Candidates)**
China Association for Science and Technology. Funding: CNY 40,000. Role: Host.

Dec. 2025–Dec. 2026 **CSC Joint PhD Training Program**
China Scholarship Council. Funding: USD 14,400. Role: Host.

Jan. 2024–Dec. 2025 **Fundamental Research Fund for the Central Universities Project**
Funding: CNY 30,000. Role: Host.

AWARDS AND HONORS

Sep. 2024–Sep. 2025 **Grand Prize Scholarship for PhD Student**
Xi'an Jiaotong University.

Sep. 2021–Sep. 2025 **Outstanding Graduate Student**
Xi'an Jiaotong University.

Sep. 2021–Sep. 2022 **National Scholarship for Graduate Students**
Xi'an Jiaotong University.

PUBLISHED AND ACCEPTED ARTICLES

Complex Analysis

[1] T. Zhang. **When do the convex hulls of zeros and critical points coincide?**
J. Math. Anal. Appl. **566** (2027), no. 1, Paper No. 131002. [DOI]

[2] T. Zhang. **Angle duality and a gap principle for convex combinations of incomplete polynomials on the unit circle**.
Canad. Math. Bull. (to appear). [arXiv:2601.20748]

[3] T. Zhang. **Distributions of zeros of a complex polynomial**.
J. Math. Anal. Appl. **561** (2026), no. 1, Art. No. 130609. [DOI]

[4] T. Zhang. **A refinement of Pawlowski's result**.
Proc. Amer. Math. Soc. **154** (2026), no. 2, 775–782. [DOI; arXiv:2411.07105]

[5] T. Zhang. **When D-companion matrix meets incomplete polynomials**.
J. Math. Anal. Appl. **539** (2024), no. 1, Art. No. 128466. [DOI; arXiv:2310.08930]

Operator Algebra

[6] T. Zhang. **Bourin-type inequalities for τ-measurable operators in fully symmetric spaces**.
Acta Math. Sci. Ser. B (Engl. Ed.) (to appear). [arXiv:2602.00358]

[7] T. Zhang. **Proof of Audenaert–Kittaneh's conjecture**.
Israel J. Math. **273** (2026), no. 2, 811–821. [DOI; arXiv:2401.05456]

Matrix Analysis

[8] T. Zhang. **Johnson's determinantal identity for contiguous minors of Toeplitz matrices, with an accretive extension**.
J. Algebra **711** (2027), 14–25. [DOI; arXiv:2601.18977]

[9] T. Zhang. **Sharp condition-number bounds for growth factors of Higham matrices in Gaussian elimination**.
Math. Comp. (to appear). [arXiv:2604.23024]

[10] T. Zhang. **Schur–Horn type inequalities for hyperbolic polynomials**.
Indiana Univ. Math. J. (to appear). [arXiv:2601.10602]

[11] T. Zhang. **A counterexample to the Huang–Tam conjecture and Frobenius bounds for self-commutators under λ-Aluthge transforms**.
Proc. Amer. Math. Soc. (2026), published electronically September 3, 2026. [DOI; arXiv:2603.04655]

[12] T. Zhang. **Weak majorization inequalities for the cubic and quartic coefficients of $e^{(A+B)t}$ versus $e^{At}e^{Bt}$**.
Linear Algebra Appl. (to appear). [arXiv:2601.07286]

[13] T. Zhang. **Operator symmetric moduli and sharp triangle inequalities**.
J. Lond. Math. Soc. (2) **114** (2026), no. 2, Art. No. e70672. [DOI; arXiv:2603.01046]

[14] T. Zhang. **Three-factor positive results and counterexamples for a conjectured Young-type inequality**.
Linear Algebra Appl. **750** (2026), 284–295. [DOI]

[15] T. Zhang. **The equality case in $\|f(A)\| \le f(\|A\|)$ for analytic matrix functions under strongly stable norms**.
Linear Algebra Appl. **750** (2026), 154–162. [DOI]

[16] T. Zhang. **A new proof of Lee's conjecture on the Frobenius norm via the matrix Cauchy–Schwarz inequality**.
Linear Algebra Appl. **725** (2025), 355–358. [DOI; arXiv:2507.02684]

[17] F. Nan and T. Zhang. **An improvement and a generalization of Rotfel'd type inequalities for sectorial matrices**.
Linear Multilinear Algebra **73** (2025), no. 4, 675–683. [DOI; arXiv:2404.12849]

[18] M. Lin, J. Zhang, and T. Zhang. **A lower bound for the spectral radius of complex matrices**.
Linear Algebra Appl. **656** (2023), 1–8. [DOI]

[19] M. Lin and T. Zhang. **On an upper bound for the eigenvalues of the solution of the continuous algebraic Riccati equation**.
Appl. Math. Lett. **132** (2022), Art. No. 108194. [DOI]

PREPRINTS AND MANUSCRIPTS

Complex Analysis

[1] A. Eremenko, T. W. Ng, and T. Zhang. **Entire solutions of higher order Briot–Bouquet equations**.
Preprint, 2026. [arXiv:2610.20438]

[2] T. Zhang. **The Bloch constant is strictly smaller than the Ahlfors–Grunsky value**.
Preprint, 2026. [arXiv:2610.16808]

[3] T. Zhang. **Borcea's 2-variance conjecture**.
Preprint, 2026. [arXiv:2610.02035]

[4] T. Zhang. **Baernstein's quasi-norm monotonicity conjecture for polynomials with unimodular zero**. Preprint, 2026. [arXiv:2610.02009]

[5] A. Eremenko and T. Zhang. **Holomorphic curves of finite lower order with few inflection points**. Preprint, 2026. [arXiv:2609.38032]

[6] A. Eremenko and T. Zhang. **A counterexample to Fuchs's conjecture**. Preprint, 2026. [arXiv:2609.28443]

[7] A. Eremenko, Z. Xu, and T. Zhang. **The modified Cartan conjecture**. Preprint, 2026. [arXiv:2609.23674]

[8] T. Zhang. **Sendov's conjecture holds for every degree $n \geq 10^{200000}$**. Preprint, 2026. [arXiv:2609.20256]

[9] T. Zhang. **Beyond Sendov's conjecture: the quadratic Tang–Zhang inequality**. Preprint, 2026. [arXiv:2609.19126]

[10] Y. He and T. Zhang. **A counterexample to Nevanlinna's century-old half-plane problem**. Preprint, 2026. [arXiv:2608.24829]

[11] J. Lei and T. Zhang. **Proof of the Agler–McCarthy entropy conjecture**. Preprint, 2026. [arXiv:2605.03949]

[12] Y. He, Q. Tang, and T. Zhang. **Optimal universal growth for integral means of normalized logarithmic derivatives in the Carathéodory class**. Preprint, 2026. [arXiv:2603.23734]

[13] Y. He, Q. Tang, and T. Zhang. **A solution to a problem of Rubel on two-parameter normal families of entire functions**. Preprint, 2026. [arXiv:2603.20883]

[14] Q. Tang and T. Zhang. **Sharp Schoenberg type inequalities and the de Bruin–Sharma problem**. Preprint, 2025. [arXiv:2508.10341]

Operator Algebra

[15] Z. Xu and T. Zhang. **Sharp tangent inequalities in noncommutative L_p-spaces**. Preprint, 2026. [arXiv:2609.16202]

[16] Y. He, Q. Tang, Z. Xu, and T. Zhang. **The nonseparable case of Kadison's problem on orthonormal bases of unitaries for type II_1 factors**. Preprint, 2026. [arXiv:2608.18405]

[17] T. Zhang. **On the embedding rigidity problem for uniformly locally finite coarse spaces**. Preprint, 2026. [arXiv:2607.16949]

[18] T. Zhang. **Isomorphism rigidity of uniform Roe algebras over arbitrary uniformly locally finite coarse spaces**. Preprint, 2026. [arXiv:2607.15096]

[19] Y. He, Q. Tang, and T. Zhang. **The separable case of Kadison's problem on orthonormal bases of unitaries for type II_1 factors**. Preprint, 2026. [arXiv:2605.15006]

[20] L. Molnár and T. Zhang. **On interrelations among different versions of a Heron type mean and commutativity in C^*-algebras**. Preprint, 2026. [arXiv:2602.14123]

[21] L. Molnár and T. Zhang. **Kadison's theorem on linear order isomorphisms of C^*-algebras holds without the additivity assumption**. Manuscript, 2026. [PDF]

[22] T. Zhang. **Trace arithmetic–κ_p inequality**. Preprint, 2026. [arXiv:2602.11922]

[23] T. Zhang. **Metric properties: from S-divergence to quantum Jensen divergence**. Preprint, 2026. [arXiv:2602.08164]

[24] T. Zhang. **Clarkson–McCarthy type inequalities, Part I: ℓ_p–ℓ_p and ℓ_q–ℓ_p Schatten p-estimates**. Preprint, 2024. [arXiv:2410.21961]

Matrix Analysis

[25] T. Zhang. **The A-optimal design problem and the S-matrix conjecture**. Preprint, 2026. [arXiv:2609.20306]

[26] T. Zhang. **Plank theorems, Gaussian probabilistic estimates and Rump's 100 Euro conjecture**. Preprint, 2026. [arXiv:2603.07423]

[27] T. Zhang. **An operator triangle inequality for the quadratic symmetric modulus**. Preprint, 2026. [arXiv:2602.01463]

[28] Z. Xu and T. Zhang. **Sharp perturbation bounds on the Frobenius norm of subunitary and positive polar factor**. Preprint, 2025. [arXiv:2507.14940]

Group Theory

[29] J. Lei and T. Zhang. **Colored Stallings graphs and counterexamples to Stallings equalizer conjecture**. Preprint, 2026. [arXiv:2604.24502]

[30] J. Lei and T. Zhang. **Nonscalar non-commutators in elementary linear groups**. Preprint, 2026. [PDF]

INVITED TALKS

Dec. 2024 — **Joint Symposium by Five Schools of Mathematics**
Renmin University of China, Xi'an Jiaotong University, Beijing Institute of Technology, Shandong University, and Beijing Normal University. Online.

Sep. 2024 — **2nd Symposium on Frontiers in Mathematics for PhD Students**
Central China Normal University.

Oct. 2023 — **1st Symposium on Frontiers in Mathematics for PhD Students**
Peking University.

TEACHING ASSISTANT EXPERIENCE

Sep. 2023–Jan. 2024 — **Complex Analysis, Honors Mathematics Program**
Xi'an Jiaotong University.

Sep. 2021–Jun. 2022 — **Linear Algebra and Analytic Geometry, Undergraduate Courses**
Xi'an Jiaotong University.